

Measuring What Drives Citation in AI Search Engines

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The Problem

A single measurement tells you almost nothing about a page's true visibility in AI search

- **AI search engines** (ChatGPT, Perplexity, Claude) now answer questions directly and cite only a few sources
- **Traditional SEO** has Google Search Console, you can see exactly how your pages rank
- **GEO has no equivalent tool** no visibility dashboard, no ranking report
- **Worse:** The same prompt produces different answers each time
 - Schulte et al. (2026): only **34–42% source overlap** between repeated runs
 - My preliminary data: only **14% of retrieved URLs stable** across 3 runs

Data Collected And a Preliminary Finding

Three Datasets

Dataset	Scale	What It Contains
AI citations	1,728 citations across 898 domains	OpenAI, Perplexity, Claude × 100 prompts
Google comparison	765 top-10 results	Non-cited control group
Controlled experiments	5 documents × 3 engines × 10 runs	Manually specified document queue

Plan and Contribution

The Three Tests

Test	Question	Method
Test 1	How stable is retrieval and citation across runs?	Jaccard + RBO across 5–8 runs per engine
Test 2	Does adding/removing a source change citation odds?	Plackett-Luce model + chi-square test
Test 3	Can citation be predicted from rank and content?	Logistic regression; rank vs. features

Expected Contribution and Timeline

- **Reproducible measurement framework** for AI search visibility across three deployed engines
- **Empirical evidence** on how much citation depends on content vs. retrieval rank vs. randomness
- **First empirical test** of whether LLM citation satisfies Independence of Irrelevant Alternatives

Timeline

- **Weeks 1–2:** Complete data collection
- **Weeks 3–4:** Test 1 (stability)
- **Weeks 5–7:** Test 2 (IIA)
- **Weeks 8–9:** Test 3 (prediction)
- **Weeks 10–12:** Write-up

Risk Mitigation

If time is tight, I'll complete IIA testing fully on **two engines** instead of three.